Fiscal Year:	FY 2021	**Task Last Updated:**	FY 12/22/2020
PI Name:	Britten, Richard Ph.D.		
Project Title:	Changes in the Neuroproteome Associated with HZE-Induced Impairment of Cognition		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	23507-1607	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Space Radiobiology NNJ13ZSA001N
Start Date:	02/27/2014	**End Date:**	04/30/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Element change from SR to HFBP per Human Research Roadmap information dtd July 2019 (Ed., 1/6/2020) NOTE: Extended to 4/30/2023 per NSSC information (Ed., 12/30/2020) NOTE: Extended to 12/31/2020 per NSSC information (Ed., 3/12/19) NOTE: Extended to 12/31/2018 per S. Monk/SR/LaRC (Ed., 1/11/18)		
Key Personnel Changes/Previous PI:	December 2019 report: Dr. Douglas Wallace at Northern Illinois University (NIU) is now a CoInvestigator. Dec 2016: Drs. Semmes and Dutta were removed from the project; proteomic analysis is now being conducted at UTMB (University of Texas Medical Branch) as contract work.		
COI Name (Institution):	Wallace, Douglas Ph.D. (Northern Illinois University)		
Grant/Contract No.:	NNX14AE73G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	One of NASA's biggest concerns about the long-term health of astronauts who go on deep space missions is the impact that High Z, High Energy (HZE) particles have on brain function (neurocognition). Data from our laboratory and others suggests that there is significant impairment of certain neurocognitive tasks (spatial memory and Executive function-attentional set shifting) following exposure to low HZE doses. The goal of this application is to determine the Threshold dose for the induction of HZE-induced spatial memory impairments (HISMI) or Attentional Set Shifting Impairments (HIASSI) following exposure to 56Fe, 48Ti, and 28Si particles. The proposed studies will also identify the changes in the proteome of the brain (neuroproteome) of rats that differ in their susceptibility to HISMI and HIASSI, which will provide further insight into the factors that lead to HISMI/HIASSI and perhaps more importantly, that prevent its emergence. Our underlying hypothesis is that HISMI and HIASSI arise as the direct result of HZE-induced changes in the neuroproteome. We also hypothesize that exposure to HZE species that have different track structures will result in different mechanisms of HZE-induced cognitive impairment (HICI). Collectively, these studies will give some insight into the underlying cause for HISMI and HIASSI. Our studies will thus address CNS (Central Nervous System) Gaps 1, 2, and 6, and we shall specifically focus on the following aims: Aim 1. Determine the Threshold dose for the induction of HISMI and HIASSI following exposure to 56Fe, 48Ti, and 28Si particles when delivered as a single dose. Aim 2. Identify changes in the neuroproteome that are associated with susceptibility or resistance to developing HISMI and HIASSI following exposure to 56Fe particles. Aim 3. Determine the mechanism of HISMI and HIASSI induced by HZE particles of differing LET (linear energy transfer). In Aim 1, socially mature (~6 month old) male Wistar rats will be irradiated with 56Fe, 48Ti, and 28Si particles (with incident energies of 600 MeV/nucleon). Rats will receive whole body HZE irradiation (< 15 cGy), and HISMI and HIASSI will be assessed at 3 months post irradiation. In Aim 2 and 3, the composition of the neuroproteome (hippocampus and selected regions of the prefrontal cortex) of irradiated rats that have “normal” cognitive performance or have developed HISMI or HIASSI will be established using an unbiased proteomic profiling approach. We shall use a label free differential protein profiling workflow on the Q-Exactive Orbitrap mass spectrometer. These studies will give considerable insight into the underlying cause for HZE-induced neurocognitive failure. The proposed studies will continue to define the minimum dose of HZE particles that will induce HISMI and HIASSI. Moreover, our studies will provide considerable insight into the underlying mechanism of HICI, and will identify prognostic biomarkers that could be translated to human studies to monitor the emergence of HICI. These studies may also help to develop appropriate countermeasures and help identify sensitive individuals, so that NASA’s medical staff can implement appropriate countermeasures to protect these at risk individuals. Supplemental studies (in December 2019 report) This study will provide information on the robustness of single-exposure experiments to predict the impact of repeated episodic radiation exposures (such as will be encountered on the mission to Mars) on neurocognition. This study will test the hypothesis that episodic space radiation (SR) exposure will result in more severe neurocognitive deficits than single, or multiple daily SR doses. This study will utilize both male and female rats, and two different radiation regimens incorporating 4He ions and the 5-ion galactic cosmic ray simulation (GCRsim) beam. Performance in one executive function task (Attentional Set shifting-ATSET) will be assessed after a single exposure (He or GCRsim) and after a second exposure (~6 months later) to the 5-ion GCRsim beam.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These studies will give considerable insight into the underlying cause for Space radiation (SR)-induced neurocognitive impairment (SICI). The proposed studies will continue to define the minimum dose of SR particles that will impair cognitive flexibility (Attentional Set shifting and Unconstrained cognitive flexibility) performance. Importantly both of our cognitive flexibility tasks are homologs of tasks used in clinical testing of humans. Our studies will model the impact that single and repeated episodic exposure to SR has on neurocognitive performance and fine motor skills. Moreover, our studies will provide considerable insight into the underlying mechanism of SICI, and will identify prognostic biomarkers that could be translated to human studies to monitor the emergence of SICI. These studies may also help to develop appropriate countermeasures and help identify sensitive individuals, so that NASA’s medical staff can implement appropriate countermeasures to protect these at risk individuals.
	Project Objectives 1. Identify the lowest dose of space radiation (SR) that results in Attentional Set Shifting (ATSET) impairment. 2. Determine if there are LET-specific mechanisms of ATSET impairment. 3. Identify changes in the neuroproteome that reflect the cognitive performance status of SR-exposed animals. Supplemental studies 4. Establish the impact that that re-irradiation with 10 cGy of simplified (5-ion) CGRsim beam has on the ATSET performance of male and female Wistar rats that maintained a functional ATSET performance after exposure to 10 cGy of either He or GCRsim. Project Approach To better simulate the “clinical reality,” adult (proven breeder) rats that have been pre-selected for good attentional set shifting (ATSET) performance, and who have been maintained on an exercise regimen are used in this study.

Task Progress:	Rats are exposed (primarily a single exposure) to <15 cGy of SR ions (incident energy <1000 MeV/n) and ATSET performance re-established at 3 months post exposure. After completion of the ATSET test, the rats are then tested in the Unrestrained Cognitive flexibility (UCFlex) assay, which requires the rats to complete a new task (where the food reward is no longer present in either reward pot, as it was for all seven stages of the ATSET; instead the reward is located in a third location that the rat had limited experience with) that requires the rat to develop a novel solution to obtain the food reward. Thus, the UCFlex version of the ATSET provides the ability to measure both constrained and unconstrained cognitive flexibility within individual rats. Importantly both cognitive flexibility tasks are homologs of tasks used in clinical testing of humans. Through a new collaboration with Dr. Wallace at Northern Illinois University, we have shipped our rats to NIU, where Dr. Wallace's team has determined whether there are sensorimotor defects induced by SR, and if any observed changes are co-incidental with changes in ATSET/UCF performance. Specifically, fine motor skills have been assessed using the string-pull assay. Regions of the cortex, basal forebrain, hippocampus (that regulated certain paradigms with the ATSET test), and temporal lobe (that probably regulates UCFlex performance) are recovered and subjected to proteomic analysis to identify some of the processes that may be responsible for the SR-induced impairment of ATSET. The supplemental studies involve returning the rats that have maintained good ATSET performance after SR (10 cGy 4He ions or the 5-ion GCRsim beam) to Brookhaven National Laboratory (BNL) where they receive a second dose of 10 cGy of the 5-ion GCRsim. ATSET performance is then reassessed at 3 months after the second exposure. Research Highlights from this reporting period. • The COVID pandemic severely interrupted work on this project for the majority of this reporting period; however, progress was made in several areas. • We published the first study to report that exposure to mission relevant space radiation doses (5 cGy of Si) results in fine motor skill loss, as measured using the string pull assay (Blackwell et al., 2020). The loss of performance in the string pull assay was evident at 3 months post exposure, in rats that had no obvious loss of neurocognitive performance. Ongoing studies are determining how rapidly the impairment of fine motor skills occurs after radiation exposure. • We have determined that exposure to =1 cGy 400 MeV/n 4He ions results in a significant loss of ATSET performance, and also social withdrawal in male rats. This is the first study to demonstrate space radiation-induced social withdrawal in freely-interacting rodents. Some individuals in confinement studies such as the Mars 500 study (Basner et al., 2014) did socially withdraw, so there is a possibility that SR exposure could exacerbate such withdrawals. • We have demonstrated for the first time that there is a marked dose responsiveness of UCFlex impairment following He exposure, in stark contrast to other previous studies with neutrons and Si ions. Once again there is no concomitant loss of ATSET and UCFlex performance in individual rats. • We have demonstrated that exposure to 10 cGy GCRsim results in impaired ATSET performance in female rats. These data suggest that there is no panoramic resilience of female rodents to the impact of SR on neurocognition, as implicated by the mouse/ NOR studies. • We have demonstrated that exposure to 10 cGy GCRsim leads to a significant reduction in the ability of female to perform pattern separation and avoid anterograde interference in rats that had no obvious loss of ATSET performance. These studies provide further evidence that reliance on a sole measure of neurocognitive performance may lead to significant underestimate of the impact of SR on cognition, and suggest that as task complexity (cognitive loading) increases so does the likelihood of SR-induced performance impairment. • In collaboration with NASA GeneLab we have determined that exposure to 5 cGy Si results in significant changes in the proteomic composition of the prefrontal cortex, i.e., identified protein biomarkers for space radiation exposure and further subset that dissociated rats that have impaired ATSET performance from those that have apparently normal performance, e.g., the mitochondrial oxidative phosphorylation pathway. Reference: Basner M, Dinges DF, Mollicone DJ, Savelev I, Ecker AJ, Di Antonio A, Jones CW, Hyder EC, Kan K, Morukov BV, Sutton JP. "Psychological and behavioral changes during confinement in a 520-day simulated interplanetary mission to Mars." PLoS One. 2014 Mar 27;9(3):e93298.

Bibliography Type:	Description: (Last Updated: 03/31/2026)
Articles in Peer-reviewed Journals	Britten RA, Duncan VD, Fesshaye A, Rudobek E, Nelson GA, Vlkolinsky R. "Altered cognitive flexibility and synaptic plasticity in the rat prefrontal cortex after exposure to low (< or =15 cGy) doses of 28Si radiation." Radiat Res. 2020 Mar;193(3):223-35. https://doi.org/10.1667/RR15458.1 ; PMID: 32011211 , Mar-2020
Articles in Peer-reviewed Journals	Britten RA, Fesshaye A, Duncan VD, Wellman LL, Sanford LD. "Sleep fragmentation exacerbates executive function impairments induced by low doses of Si ions." Radiat Res. 2020 Aug 1;194(2):116-23. https://doi.org/10.1667/RADE-20-00080.1 ; PMID: 32845991 , Aug-2020
Articles in Peer-reviewed Journals	Blackwell AA, Schell BD, Osterlund Oltmanns JR, Whishaw IQ, Ton ST, Adamczyk NS, Kartje GL, Britten RA, Wallace DG. "Skilled movement and posture deficits in rat string-pulling behavior following low dose space radiation (28Si) exposure." Behav Brain Res. 2021 Feb 26;400:113010. Epub 2020 Nov 10. https://doi.org/10.1016/j.bbr.2020.113010 ; PMID: 33181183 , Feb-2021